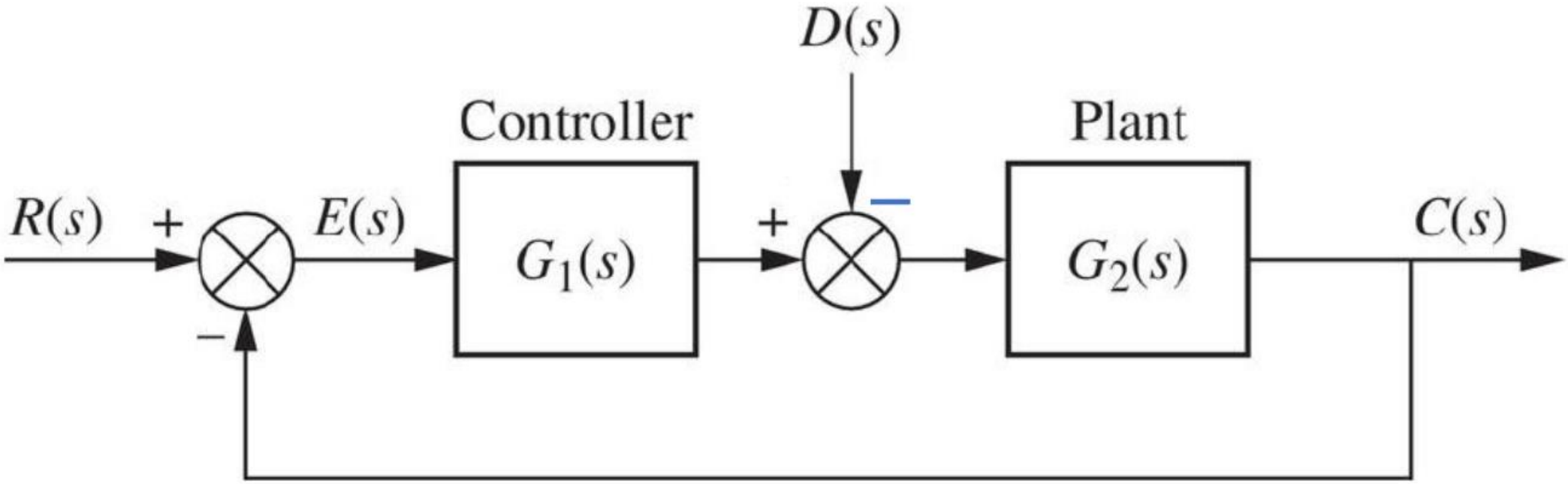
Steady State Error_2

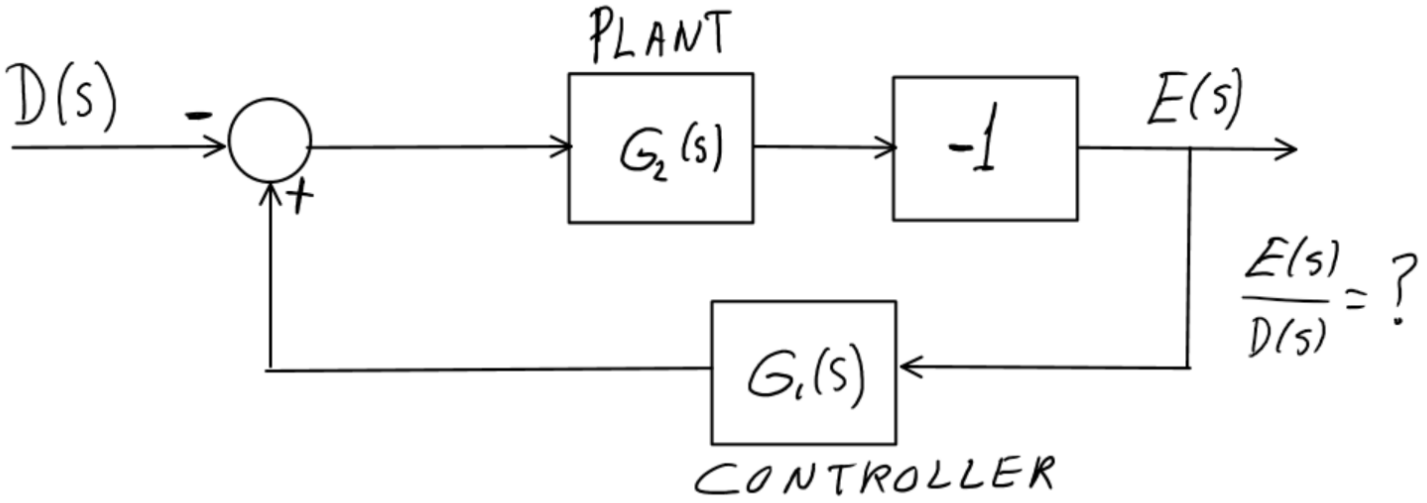
(Due to disturbance)

Block Diagram with Disturbance



Calculating E(s) due to D(s)

- The error E(s) due to the disturbance is calculated by making R(s) = 0
- Then, our block diagram can be transformed into



Calculating E(s) due to D(s)

- Applying the known closed loop formula:

$$\frac{E(s)}{D(s)} = (-1)\frac{G_2(s)(-1)}{1 - G_2(s)(-1)G_1(s)} = \frac{G_2(s)}{1 + G_1(s)G_2(s)}$$

- Assume $G(s) = G_1(s)G_2(s)$, then

$$E(s) = \frac{G_2(s)}{1 + G(s)}D(s)$$

- Remember that $G_1(s)$ plays the role of the controller in this case.

Steady State Error Due to the Disturbance

- For the steady state error calculation, we need to apply the Final Value Theorem (assuming all poles are in LHP)

$$e_{ss} = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \frac{sG_2(s)}{1+G(s)} D(s)$$

- Where $G(s) = G_1(s)G_2(s)$
- Now we need to use the different inputs (step, ramp and parabola) for $D(s)$

Step Disturbance $D(s) = d/s$

$$e_{ss} = \lim_{s\to 0} s\left(\frac{d}{s}\right)\frac{G_2(s)}{(1+G_1(s)G_2(s))} = \lim_{s\to 0} d\,\frac{G_2(s)}{1+G_1(s)G_2(s)}$$

- If $G(s) = G_1(s)G_2$(s) is type 0 (which means that both G_1 and G_2 are type 0), we have

$$e_{ss} = \frac{dG_2(s)}{1+G_1(s)G_2(s)} = \frac{dK_{p2}}{1+K_{p1}K_{p2}}$$

- Where $K_{p2} = \lim_{s\to 0} G_2(s)$ and $K_{p1} = \lim_{s\to 0} G_1(s)$

Step Disturbance $D(s) = d/s$

- If G_1(s) is type 0 and $G_2(s)$ is type 1 or higher (which means that $G = G_1 G_2 \to \infty$ as $s \to 0$), we have

$$e_{ss} = \lim_{s\to 0} d \frac{G_2(s)}{1 + G_1(s)G_2(s)} = \lim_{s\to 0} d \frac{G_2(s)}{G_1(s)G_2(s)} = \frac{d}{\lim_{s\to 0} G_1(s)} = \frac{d}{K_{p1}}$$

- If G_1(s) is type 1 or higher regardless of the type of $G_2(s)$, we have

$$e_{ss} = \lim_{s\to 0} d \frac{G_2(s)}{1+G_1(s)G_2(s)} = \lim_{s\to 0} d \frac{G_2(s)}{G_1(s)G_2(s)} = \frac{d}{\lim_{s\to 0} G_1(s)} = \frac{d}{\infty} = 0$$

Ramp Disturbance $D(s) = d/s^2$

- In this case we have

$$e_{ss} = \lim_{s\to 0} s\left(\frac{d}{s^2}\right)\frac{G_2(s)}{(1+G_1(s)G_2(s))} = \lim_{s\to 0}\frac{d}{s}\frac{G_2(s)}{(1+G_1(s)G_2(s))}$$

- Simplifying

$$e_{ss} = \lim_{s\to 0}\frac{dG_2(s)}{sG_1(s)G_2(s)} = \lim_{s\to 0}\frac{d}{sG_1(s)} = \frac{d}{\lim_{s\to 0} sG_1(s)} = \frac{d}{K_{v1}}$$

Ramp Disturbance $D(s) = d/s^2$

- Therefore, if G_1 is type 0 ($K_{v1} = 0$), we have $e_{ss} = \infty$

- If G_1 is type 1 ($K_{v1} =$ a number diferent from zero), the error is finite given by

$$e_{ss} = \frac{d}{K_{v1}}$$

- If G_1 is type 2 or higher. We have $e_{ss} = 0$

Parabola Disturbance $D(s) = d/s^3$

- In this case we have

$$e_{ss} = \lim_{s\to 0} s\left(\frac{d}{s^3}\right)\frac{G_2(s)}{1+G_1(s)G_2(s)} = \lim_{s\to 0}\frac{d}{s^2}\frac{G_2(s)}{(1+G_1(s)G_2(s))}$$

- Simplifying

$$e_{ss} = \lim_{s\to 0}\frac{dG_2(s)}{s^2G_1(s)G_2(s)} = \lim_{s\to 0}\frac{d}{s^2G_1(s)} = \frac{d}{\lim_{s\to 0} s^2G_1(s)} = \frac{d}{K_{a1}}$$

Parabola Disturbance $D(s) = d/s^3$

- Therefore, if G_1 is type 0 or 1 $(K_{a1} = 0)$, we have $e_{ss} = \infty$

- If G_1 is type 2 ($K_{a1} =$ a number diferent from zero), the error is finite given by

$$e_{ss} = \frac{d}{K_{a1}}$$

- If G_1 is type 3 or higher. We have $e_{ss} = 0$

Summary

- Notice that the error due to disturbance depends mostly on G_1, which plays the role of the controller.
- The final Summary is in document *Steady State Error Summary.pdf,* which concludes on the total error: error due to the reference (main input) plus error due to the disturbance.